In addition, we have

$$\begin{aligned} \pi_{2n+1}(\varphi) = \nu_V \mathcal{S}_{2n+1}(\varphi) &\leqslant \epsilon\, \nu_V(\psi) + c \\ \pi_{2n}(\psi) = \lambda_U \mathcal{S}_{2n}(\psi) &\leqslant \epsilon\, \lambda_U(\varphi) + c \quad \text{with } (\epsilon, c) \text{ as in (4.13).} \end{aligned} \tag{4.15}$$

Theorem 4.6. *We have the following:*

- *Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ holds as soon as condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ is satisfied and the estimates (4.12) hold for some $n_0 \geqslant 1$ and some parameters $\rho \in [0,1[$ and $\kappa > 0$.*
- *When $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ is satisfied there exist parameters $\kappa \geqslant 0$ and $\rho \in [0,1[$ that only depend on the parameters (ϵ, c) in (4.13) and on the function $\varepsilon(r)$ in (4.14) and such that, for any $n \geqslant n_0$,*

$$\chi_{\varphi_\kappa,\psi_\kappa}(\mathcal{S}_{2n}) \vee \chi_{\psi_\kappa,\varphi_\kappa}(\mathcal{S}_{2n+1}) < \rho, \quad \textit{where} \quad (\varphi_\kappa, \psi_\kappa) := (1+\kappa\varphi, 1+\kappa\psi).$$

A proof of Theorem 4.6 is provided in Appendix C (on page 38).

Let us emphasize that Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ is stronger than Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$. In particular, the Gibbs-loop transitions $\mathcal{S}_n^\circ$ may be stable even in situations where the Sinkhorn transitions $\mathcal{S}_n$ are unstable. Intuitively, Gibbs-loop transitions alternate between forward and backward steps that steer the dynamics toward their invariant measures (2.8), whereas the Sinkhorn transitions in (2.2) asymptotically transport one target marginal measure to the other.

As an illustration, Lemma 4.9 in [2] (see also Section A) shows that the Gibbs-loop transitions $\mathcal{S}_n^\circ$ of the Gaussian Sinkhorn algorithm are always stable. By contrast, in the setting of regularized Gaussian entropic problems over a finite time horizon t (see Section 3.4 in [2]), the Sinkhorn transitions $\mathcal{S}_n$ may become unstable when the time horizon is not sufficiently large; see Remarks 3.9 and 4.6 in [2].

4.4. A Lyapunov stability theorem.

4.4.1. *Regularity conditions.* This section presents some sufficient conditions for $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$, in terms of the potential functions (U, V, W). These simple conditions enable us to obtain geometric rates for the convergence of the Sinkhorn iterations for a variety of models, including polynomial growth models (Section 5.1) and heavy tailed models (Section 5.2), as well as boundary states models including Beta and Weibull distributions (Section 5.3 and Section 5.4). Illustrations in the context of Linear-Gaussian models and finite (non necessarily Gaussian) mixture marginals are also discussed respectively in Section 5.8 and Section 5.9.

Let us assume that U, V, W are locally bounded and $e^{-W} \in \mathbb{L}_\infty(\lambda \otimes \nu)$. In addition, U, V have compact sub-level sets. These conditions ensure there exists some $W_\star \in \mathbb{R}$ such that

$$W(x,y) \geqslant W_\star \quad \text{for } (\lambda \otimes \nu)\text{-almost every } (x,y) \in (\mathbb{X} \times \mathbb{Y}). \tag{4.16}$$

In addition, (U, V) are bounded below by a real number so that e^{-U} and e^{-V} are uniformly bounded. This excludes probability densities that tends to infinity at boundary states, such that some classes of Weibull or the Beta densities for some shape parameters. Also note that for any $\delta > 0$ we have

$$(e^{\delta U}, e^{\delta V}) \in (\mathcal{B}_\infty(\mathbb{X}) \times \mathcal{B}_\infty(\mathbb{Y})).$$

Consider the following conditions:

Condition $\mathcal{H}_\delta(U, V)$: *There is some $\delta \in]0,1[$ such that the functions $(\mathcal{U}_\delta, \mathcal{V}_\delta)$ defined in (2.13) are locally bounded with compact sub-level sets and we have*

$$\lambda_{(1-\delta)U}\left(e^{W^V}\right) \vee \nu_{(1-\delta)V}\left(e^{W_U}\right) < \infty. \tag{4.17}$$

Note that (4.16) and (4.17) together imply the inequalities

$$\lambda_{(1-\delta)U}(1) \vee \nu_{(1-\delta)V}(1) \leqslant e^{-W_\star} \left(\lambda_{(1-\delta)U}\left(e^{W^V}\right) \vee \nu_{(1-\delta)V}\left(e^{W_U}\right)\right) < \infty.$$

This yields the exponential moment estimates

$$\mathcal{H}_\delta(U,V) \Longrightarrow \lambda_{(1-\delta)U}(1) = \lambda_U(\exp(\delta U)) < \infty \quad \text{and} \quad \nu_{(1-\delta)V}(1) = \nu_V(\exp(\delta V)).$$

In particular all absolute moments $\lambda_U(|U|^p) \vee \nu_V(|V|^p) < \infty$, with $p \geqslant 0$ are finite. Given the reference measure $\mathcal{P}_0 = (\lambda_U \times \mathcal{Q})$ also have

$$\mathcal{H}_\delta(U,V) \Longrightarrow \mathrm{Ent}((\lambda_U \otimes \nu_V) \mid \mathcal{P}_0) = (\lambda_U \otimes \nu_V)(W) - \nu_V(V) < \infty. \tag{4.18}$$

To check this claim, note that

$$\begin{aligned}(\lambda_U \otimes \nu_V)(W) &= \lambda_U(W^V) = W_\star + \lambda_U(W^V - W_\star)\\ &\leqslant W_\star + \lambda_U(e^{W^V - W_\star}) = W_\star + e^{-W_\star}\ \lambda_{(1-\delta)U}(e^{-\mathcal{U}_\delta}).\end{aligned}$$

On the other hand, for any $r > 0$ we have

$$\begin{aligned}\lambda_{(1-\delta)U}(e^{-\mathcal{U}_\delta}) &\leqslant e^{-r}\ \lambda_{(1-\delta)U}(1) + \lambda_{(1-\delta)U}(e^{-\mathcal{U}_\delta}\ 1_{\mathcal{U}_\delta \leqslant r})\\ &\leqslant \lambda_{(1-\delta)U}(1)\ \left(e^{-r} + \exp\left(\sup_{\mathcal{U}_\delta(x)\leqslant r} |\mathcal{U}_\delta(x)|\right)\right) \Longrightarrow (4.18).\end{aligned}$$

Condition $\mathcal{H}'_\delta(U,V)$*: There is some* $\delta \in]0,1/2[$ *such that* $\mathcal{U}_\delta$ *and* $\mathcal{V}_\delta$ *are locally bounded with compact sub-level sets and we have*

$$\lambda_{(1-2\delta)U}(1) \vee \nu_{(1-2\delta)V}(1) < \infty. \tag{4.19}$$

Let us remark that

$$\mathcal{H}'_\delta(U,V) \Longrightarrow \mathcal{H}_\delta(U,V). \tag{4.20}$$

To verify this claim, note that the regularity condition on $(\mathcal{U}_\delta, \mathcal{V}_\delta)$ ensures that

$$(e^{\mathcal{U}_\delta}, e^{\mathcal{V}_\delta}) \in (\mathcal{B}_\infty(\mathbb{X}) \times \mathcal{B}_\infty(\mathbb{Y})).$$

When $\mathcal{U}_\delta$ is locally bounded with compact sub-level sets $C_{\mathcal{U}}(r) := \{\mathcal{U}_\delta \leqslant r\}$ for any $0 < \delta < 1/2$ we have

$$\begin{aligned}\lambda_{(1-\delta)U}\left(e^{W^V}\right) &= \lambda_{(1-2\delta)U}\left(e^{-\mathcal{U}_\delta}\right)\\ &\leqslant \left(e^{-r} + e^{c_\delta(r)}\right)\lambda_{(1-2\delta)U}(1) \quad \text{with} \quad c_\delta(r) := \sup_{x\in C_{\mathcal{U}}(r)} |\mathcal{U}_\delta(x)|.\end{aligned}$$

In this context, we readily see that

$$\lambda_{(1-2\delta)U}(1) \vee \nu_{(1-2\delta)V}(1) < \infty \Longrightarrow (4.17). \tag{4.21}$$

This ends the proof of (4.20).

Remark 4.7. *Whenever* $\|W\| < \infty$ *we have* $\|W_U\| \vee \|W^V\| < \infty$*. In this case* $\mathcal{H}_\delta(U,V)$ *is satisfied as soon as there is some* $\delta \in]0,1[$ *such that*

$$\lambda_{(1-\delta)U}(1) \vee \nu_{(1-\delta)V}(1) < \infty. \tag{4.22}$$

4.4.2. *A Lyapunov stability theorem.*

Theorem 4.8. *If Condition* $\mathcal{H}_\delta(U,V)$ *is satisfied for some* $\delta \in]0,1]$*, then Condition* $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ *is also satisfied with* $n_0 = 1$ *for the Lyapunov functions*

$$(\varphi,\psi) = (e^{\delta(U-U_\star)}, e^{\delta(V-V_\star)}) \quad \textit{with} \quad (U_\star, V_\star) := (\inf_{\mathbb{X}} U, \inf_{\mathbb{Y}} V) \in (\mathbb{R}\times\mathbb{R}). \tag{4.23}$$

In addition, there exist some parameter $\rho \in [0,1[$ *and some constant* $c > 0$ *such that*

$$\left\|\log \frac{d\pi_{2n}}{d\nu_V}\right\|_\psi \vee \left\|\log \frac{d\pi_{2n+1}}{d\lambda_U}\right\|_\varphi \leqslant c\ \rho^{2n} \quad \textit{for any } n \geqslant 1. \tag{4.24}$$

A proof of Theorem 4.8 is provided in Appendix D.4.

The above theorem combined with the estimates (4.15) ensures that

$$\begin{aligned}\sup_{n\geqslant n_0}\pi_{2n+1}(\varphi) &\leqslant \epsilon\,\nu_V(\psi)+c=\epsilon\,e^{-\delta V_\star}\,\nu_{(1-\delta)V}(1)+c\\ \sup_{n\geqslant n_0}\pi_{2n}(\psi) &\leqslant \epsilon\,\lambda_U(\varphi)+c=\epsilon\,e^{-\delta U_\star}\,\lambda_{(1-\delta)U}(1)+c\end{aligned}$$

with the parameters (ϵ,c) as in (4.13). The rescaling of the potential functions (4.23) is only made to have $\varphi\wedge\psi\geqslant 1$. Combining Theorem 4.8 with Theorem 4.6 we obtain the following corollary.

Corollary 4.9. *Assume $\mathcal{H}_\delta(U,V)$ is satisfied for some $\delta\in]0,1]$. In this situation, there exists some parameters $n_0\geqslant 0$, $\rho\in[0,1[$ and $c_1,c_2\geqslant 0$ such that for any $n\geqslant n_0$ we have*

$$\begin{aligned}\|\pi_{2n}-\nu_V\|_{tv}\leqslant c_1\,|||\pi_{2n}-\nu_V|||_\psi &\leqslant c_2\,\rho^{n-n_0}\,|||\pi_{2n_0}-\nu_V|||_\psi\\ \|\pi_{2n+1}-\lambda_U\|_{tv}\leqslant c_1\,|||\pi_{2n+1}-\lambda_U|||_\varphi &\leqslant c_2\,\rho^{n-n_0}\,|||\pi_{2n_0+1}-\lambda_U|||_\varphi,\end{aligned}\tag{4.25}$$

with the functions $(\varphi,\psi)=(e^{\delta U},e^{\delta V})$.

The r.h.s. term in (4.25) can be estimate with (4.15). Also note that

$$(4.23)\Longrightarrow \lambda_U(\varphi)=e^{-\delta U_\star}\,\lambda_{(1-\delta)U}(1)\quad\text{and}\quad \nu_V(\psi)=e^{-\delta V_\star}\,\nu_{(1-\delta)V}(1)$$

This yields the exponential decays.

Corollary 4.10. *Assume $\mathcal{H}_\delta(U,V)$ is satisfied for some $\delta\in]0,1]$. In this situation, there exists some parameters $n_0\geqslant 0$, $\rho\in[0,1[$ and $c_1,c_2\geqslant 0$ such that for any $n\geqslant n_0$ we have*

$$\begin{aligned}\|\pi_{2n}-\nu_V\|_{tv}\leqslant c_1\,|||\pi_{2n}-\nu_V|||_\psi &\leqslant c_2\,\rho^{n}\,(1+\lambda_U(\varphi)+\nu_V(\psi))\\ \|\pi_{2n+1}-\lambda_U\|_{tv}\leqslant c_1\,|||\pi_{2n+1}-\lambda_U|||_\varphi &\leqslant c_2\,\rho^{n}\,(1+\lambda_U(\varphi)+\nu_V(\psi)),\end{aligned}\tag{4.26}$$

with the functions (ψ,φ) as in Corollary 4.9.

Assume that (U,V) and (W_U,W^V) are continuous functions and we have

$$\delta U-W^V\geqslant\mathcal{U}_\delta^-\quad\text{and}\quad\delta V-W_U\geqslant\mathcal{V}_\delta^-,$$

for some functions $(\mathcal{U}_\delta^-,\mathcal{V}_\delta^-)$ with compact level sets. In this case, the functions $(\mathcal{U}_\delta,\mathcal{V}_\delta)$ are locally bounded with compact level sets. As underlined in [42], using (C.7) and Proposition D.9, the continuity condition in the above assertion can be relaxed. More precisely, Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$, Theorem 4.8 as well as Corollaries 4.9 and 4.10 hold when $(\mathcal{U}_\delta^-,\mathcal{V}_\delta^-)$ are locally bounded with compact sub-level sets. Assume we have

$$W(x,y)\leqslant a_W(x)+b_W(y)$$

for some locally bounded functions (a_W,b_W) such that $\lambda_U(a_W)\vee\nu_V(b_W)<\infty$. Note that

$$\mathcal{U}_\delta^-(x):=\delta U(x)-a_W(x)-\nu_V(b_V)\leqslant\mathcal{U}_\delta(x)\leqslant\delta U(x)-W_\star.$$

with $W_\star$ as in (4.16). This shows that $(\mathcal{U}_\delta,\mathcal{U}_\delta^-)$ and similarly $(\mathcal{V}_\delta,\mathcal{V}_\delta^-)$ with $\mathcal{V}_\delta^-:=\delta V-b_W$ are locally bounded. Moreover $(\mathcal{U}_\delta^-,\mathcal{V}_\delta^-)$ have compact compact sub-level sets if and only if $(\delta U-a_W,\delta V-b_W)$ have compact compact sub-level sets.